

Halloween Eye Cluster System

Project Description

Project Overview

This project is a large-scale outdoor Halloween animatronic display consisting of a central Master Control Unit (MCU) and up to fifteen wireless cluster controllers. Each cluster controls up to eight animatronic 'Derek' units—pairs of animated eyes concealed inside ordinary soup cans. During performances, the lids open, the eyes emerge, look around, glow with independently controlled RGB lighting, and synchronize with audio effects and other clusters to create the illusion of intelligent creatures hiding throughout the yard.

System Architecture

The system is distributed. A Raspberry Pi-based Master Control Unit orchestrates performances, schedules events, hosts a browser-based configuration interface, and communicates wirelessly with ESP32-S3 cluster controllers using ESP-NOW. Each cluster operates autonomously if communication is temporarily interrupted, allowing shows to continue reliably outdoors.

Cluster Controller

Each cluster contains an ESP32-S3 development board, an Adafruit PCA9685 16-channel servo controller, two DFPlayer Mini MP3 modules with local speakers, three PIR motion sensors, an OLED status display, DIP-switch cluster addressing, and RJ45 outputs for up to eight Derek units. Local firmware manages servos, LEDs, audio, and autonomous behaviors while receiving synchronized commands from the MCU.

Derek Unit

Each Derek contains two micro servos (pan and lift), three independently controlled WS2812 RGB LED groups (left eye, right eye, and can interior), and connects to the cluster through a single RJ45 cable carrying power and control signals. The mechanical design causes the soup can lid to open naturally as the eyes rise from inside the can.

Software

The MCU software manages performance sequencing, configuration storage, diagnostics, firmware management, and a web interface. Cluster firmware is written using the Arduino framework for the ESP32-S3 and supports synchronized animations, audio playback, lighting control, sensor reporting, and local autonomous behaviors.

Current Engineering Status

The electrical architecture, component selection, and initial KiCad PCB design have already been completed. The current engineering task is to refine the schematic, verify footprints against the selected hardware modules, optimize PCB placement and routing, resolve ERC/DRC issues, and prepare the design for manufacturing through PCBWay. The project emphasizes simplicity, reliability, modularity, and ease of outdoor deployment rather than minimizing PCB size.

Goals

The finished system should support fully synchronized theatrical performances, triggered by schedules, motion sensors, or manual control. It is intended to be expandable, maintainable, and robust enough for multiple Halloween seasons while remaining easy to assemble from commercially available modules.